

### 5.2.2 Respiration

---

Respiration is the process whereby energy stored in complex organic molecules is transferred to ATP. ATP

provides the immediate source of energy for biological processes.

---

| Learning outcomes                                                                             | Additional guidance                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i> |                                                                                                                                                                                                                                                                 |
| (a) the need for cellular respiration                                                         | To include examples of why plants, animals and microorganisms need to respire (suitable examples could include active transport and an outline of named metabolic reactions).                                                                                   |
| (b) the structure of the mitochondrion                                                        | The components of a mitochondrion including inner and outer mitochondrial membranes, cristae, matrix and mitochondrial DNA.                                                                                                                                     |
| (c) the process and site of glycolysis                                                        | To include the phosphorylation of glucose to hexose biphosphate, the splitting of hexose biphosphate into two triose phosphate molecules and further oxidation to pyruvate<br><b>AND</b><br>the production of a small yield of ATP and reduced NAD.<br><br>HSW8 |
| (d) the link reaction and its site in the cell                                                | To include the formation of Acetyl CoA by the decarboxylation of pyruvate and the reduction of NAD to reduced NAD.                                                                                                                                              |

|     |                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (e) | the process and site of the Krebs cycle                                                                                                                                    | <p>To include the formation of citrate from the acetyl group of acetyl CoA and oxaloacetate and the reconversion of citrate to oxaloacetate (names of intermediate compounds are not required)</p> <p><b>AND</b></p> <p>the importance of decarboxylation, dehydrogenation, the reduction of NAD and FAD, and substrate level phosphorylation.</p> <p>HSW8</p> |
| (f) | the importance of coenzymes in cellular respiration                                                                                                                        | With reference to NAD, FAD and coenzyme A.                                                                                                                                                                                                                                                                                                                     |
| (g) | the process and site of oxidative phosphorylation                                                                                                                          | To include the roles of electron carriers, oxygen and the mitochondrial cristae.                                                                                                                                                                                                                                                                               |
| (h) | the chemiosmotic theory                                                                                                                                                    | To include the electron transport chain, proton gradients and ATP synthase in oxidative phosphorylation and photophosphorylation.                                                                                                                                                                                                                              |
| (i) | <p>(i) the process of anaerobic respiration in eukaryotes</p> <p>(ii) practical investigations into respiration rates in yeast, under aerobic and anaerobic conditions</p> | <p>To include anaerobic respiration in mammals and yeast and the benefits of being able to respire anaerobically</p> <p><b>AND</b></p> <p>why anaerobic respiration produces a much lower yield of ATP than aerobic respiration.</p> <p><i>M0.1, M0.2, M1.1, M1.3, M2.4, M3.1, M3.2</i></p> <p><b>PAG4, PAG10, PAG11</b></p> <p>HSW3, HSW4</p>                 |
| (j) | the difference in relative energy values of carbohydrates, lipids and proteins as respiratory substrates                                                                   |                                                                                                                                                                                                                                                                                                                                                                |
| (k) | the use and interpretation of the respiratory quotient (RQ)                                                                                                                | <p>To include calculating the respiratory quotient (<i>RQ</i>) using the formula:</p> $RQ = \frac{\text{CO}_2 \text{ produced}}{\text{O}_2 \text{ consumed}}$ <p><i>M0.1, M0.2, M1.1, M1.3, M2.3</i></p>                                                                                                                                                       |

- (I) practical investigations into the effect of factors such as temperature, substrate concentration and different respiratory substrates on the rate of respiration.

To include the use of respirometers.

*M0.1, M0.2, M1.1, M1.2, M1.3, M1.6, M1.10, M2.4, M3.2, M3.3, M3.5, M3.6*

**PAG4, PAG10, PAG11**

HSW3, HSW4